

SEC 2-FA4 GROUP 3-CELLONA, BD; RICAFORT, NF-FA4

Github link:

<https://github.com/bryandioncellona/APM1110/blob/main/FA4/SEC%202-FA4%20GROUP%203-CELLONA%20C%20BD%3B%20RICAFORT%20C%20NF-FA4.Rmd>

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2. A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. For a given channel, it can be assumed that a transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0. If a signal is sent, determine the probability that:

```
binary <- data.frame(  
  Correct = c(0.95, 0.75),  
  Error    = c(0.05, 0.25),  
  Sent     = c(0.70, 0.30),  
  row.names = c("0", "1")  
)
```

binary

```
##   Correct Error Sent  
## 0    0.95  0.05  0.7  
## 1    0.75  0.25  0.3
```

(a) a 1 was received;

$$P(R = 1) = P(R = 1|T = 0) \cdot P(T = 0) + P(R = 1|T = 1) \cdot P(T = 1)$$

```
prob_get1 <- (binary["0", "Error"] * binary["0", "Sent"]) +  
             (binary["1", "Correct"] * binary["1", "Sent"])  
  
cat(round(prob_get1 * 100, 2), "%", sep = " ")
```

```
## 26%
```

(b) a 1 was transmitted given than a 1 was received.

$$P(T = 1|R = 1) = \frac{P(T=1) \cdot P(R=1|T=1)}{P(R=1)}$$

```
prob_send1_get1 <- (binary["1", "Sent"] * binary["1", "Correct"]) /  
  prob_get1  
  
cat(round(prob_send1_get1 * 100, 2), "%", sep = "")
```

```
## 86.54%
```

7. There are three employees working at an IT company: Jane, Amy, and Ava, doing 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and just 1% of Ava's work is in error. What is the overall percentage of error? If a program is found with an error, who is the most likely person to have written it?

```
# Given probabilities  
P_Jane <- 0.10  
P_Amy <- 0.30  
P_Ava <- 0.60  
  
P_E_Jane <- 0.08  
P_E_Amy <- 0.05  
P_E_Ava <- 0.01  
  
# Overall percentage of error  
P_Error <- (P_Jane * P_E_Jane) +  
  (P_Amy * P_E_Amy) +  
  (P_Ava * P_E_Ava)  
  
cat("Overall Error Probability:", P_Error, "\n")
```

```
## Overall Error Probability: 0.029
```

```
cat("Overall Error Percentage:", P_Error * 100, "%\n\n")
```

```
## Overall Error Percentage: 2.9 %
```

```
# If a program is found with an error,  
# who is the most likely person to have written it?  
employee <- c("Jane", "Amy", "Ava")  
  
# Bayes' Theorem  
P_Jane_given_Error <- (P_Jane * P_E_Jane) / P_Error  
P_Amy_given_Error <- (P_Amy * P_E_Amy) / P_Error  
P_Ava_given_Error <- (P_Ava * P_E_Ava) / P_Error
```

```

P_given_Error <- c(
  P_Jane_given_Error,
  P_Amy_given_Error,
  P_Ava_given_Error
)

# Create a table of results
E_table <- data.frame(
  Employee = employee,
  Probability = round(P_given_Error, 4),
  Percentage = round(P_given_Error * 100, 2)
)

print(E_table)

```

```

##   Employee Probability Percentage
## 1     Jane      0.2759      27.59
## 2      Amy      0.5172      51.72
## 3      Ava      0.2069      20.69

```

```

# Find the employee with the highest probability
most_likely <- which.max(P_given_Error)
most_likely_employee <- employee[most_likely]
most_likely_prob <- P_given_Error[most_likely]

cat("Most likely employee to have written the program:",
    most_likely_employee, "\n")

```

```

## Most likely employee to have written the program: Amy

```

```

cat("Probability:",
    round(most_likely_prob, 4), "\n")

```

```

## Probability: 0.5172

```

```

cat("Percentage:",
    round(most_likely_prob * 100, 2), "%\n")

```

```

## Percentage: 51.72 %

```